

UNIT 3

PROGRAMMING

FUNDAMENTALS

10TH COMPUTER SCIENCE FEDERAL BOARD

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AGENDA

- Front-end
 - Back-end
 - Forms in HTML
 - Tables in HTML
 - Animation in CSS
 - Bullet Points
 - Unit test
 - Debugging
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BASIC TERMINOLOGIES CONT....

- **Front-End Development:** It is the part of web development that deals with everything the user sees and interacts with directly in their browser. It focuses on the look, feel, and behavior of a website or web application.
- **Key Technologies:**
 - *HTML – Structure (headings, paragraphs, images)*
 - *CSS – Styling (colors, fonts, layout)*
 - *JavaScript – Interactivity (buttons, animations, forms)*
- **Examples of what front-end developers do:**
 - *Design menus, buttons, and forms*
 - *Make a website look good on mobile and desktop*
 - *Create smooth animations and pop-ups*
 - *Ensure fast loading and easy navigation*



Markup Language
Content
HTML



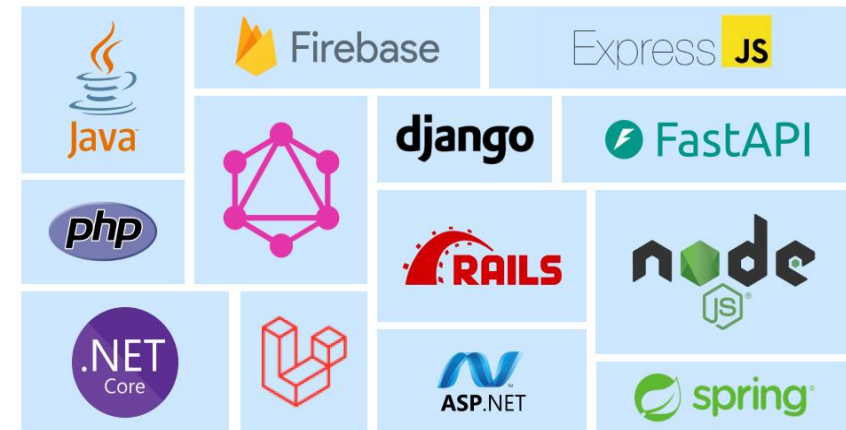
Style sheet Language
Presentation
CSS



Programming Language
Behavior
JavaScript

BASIC TERMINOLOGIES CONT....

- **Back-End Development:** It is the part of web development that deals with everything behind the scenes – the server, database, and application logic that users do not see. It makes sure the website works properly, stores data, and processes user requests.
- **Key Technologies:**
 - *Languages: Python, PHP, Java, Ruby, C#, JavaScript (Node.js)*
 - *Databases: MySQL, PostgreSQL, MongoDB (to store user data)*
 - *Servers: Apache, Nginx (to host and run the website)*
- **Examples of what back-end developers do:**
 - *Store user login details in a database*
 - *Process online payments securely*
 - *Send emails or notifications*
 - *Fetch and send data to the front-end when a user clicks a button*



FORMS IN HTML

- An HTML form is used to collect user input and send it to a server for processing. It is defined using the <form> tag, which acts as a container for various interactive controls like text fields, checkboxes, and buttons.
- Basic Form Structure

```
1 <form action="submit-endpoint"  
  method="POST">  
2   <label for="username">Username:  
    </label>  
3   <input type="text" id="username"  
    name="username" required>  
4   <label for="password">Password:  
    </label>  
5   <input type="password"  
    id="password" name="password"  
    required>  
6   <button type="submit">Log  
    In</button>  
7 </form>  
8
```

FORMS IN HTML CONT....

Input Type	Description	Example
text	Standard single-line text input	<code><input type="text"></code>
password	Masks the characters for security	<code><input type="password"></code>
email	Requires the input to look like an email address	<code><input type="email"></code>
radio	Allows selecting only one option from a group	<code><input type="radio" name="gender"></code>
checkbox	Allows selecting multiple independent options	<code><input type="checkbox"></code>
file	Allows the user to upload files	<code><input type="file"></code>
submit	A button that sends the form data to the server	<code><input type="submit"></code>
Textarea	Allow text in multiple lines	<code><textarea></code>

TABLES IN HTML

- In HTML, tables are used to arrange data into a two-dimensional grid of rows and columns. You build them using a specific set of nested tags, starting with the **<table>** container
- **<thead>**: Wraps the header rows (**<th>**) to separate them from the main content.
- **<tbody>**: Wraps the core body of data rows (**<td>**).
- **<tfoot>**: Wraps summary information or totals at the very bottom.
- **<caption>**: Adds an accessible title or description right above the table.
- **colspan**: Spans a cell horizontally across multiple columns.
- **rowspan**: Spans a cell vertically down multiple rows

```
1  <table>
2    <thead>
3      <!-- Row 1: Header Titles -->
4      <tr>
5        <th>Employee Name</th>
6        <th>Role</th>
7        <th>Joined Year</th>
8      </tr>
9    </thead>
10   <!-- Row 2: First Data Entry -->
11   <tr>
12     <td>Alice Smith</td>
13     <td>Designer</td>
14     <td>2022</td>
15   </tr>
16   <!-- Row 3: Second Data Entry -->
17   <tr>
18     <td>Bob Jones</td>
19     <td>Developer</td>
20     <td>2024</td>
21   </tr>
22 </table>
23
```

TABLES IN HTML CONT....

Essential CSS Styling

```
table {  
    width: 100%;  
    border-collapse: collapse; /* Prevents double borders between cells */  
}  
  
th, td {  
    border: 1px solid #ddd; /* Adds a light gray border */  
    padding: 8px; /* Gives text breathing room inside the cell */  
    text-align: left; /* Aligns text to the left */  
}  
  
th {  
    background-color: #f2f2f2; /* Gives headers a distinct light gray background */  
}
```

A code editor window with a dark background and light gray window controls (three circles) in the top-left corner. The code is written in a syntax-highlighted style: keywords like 'table', 'th', and 'td' are in yellow, property names are in blue, and values are in red. The code is numbered from 1 to 12 on the left side of the editor.

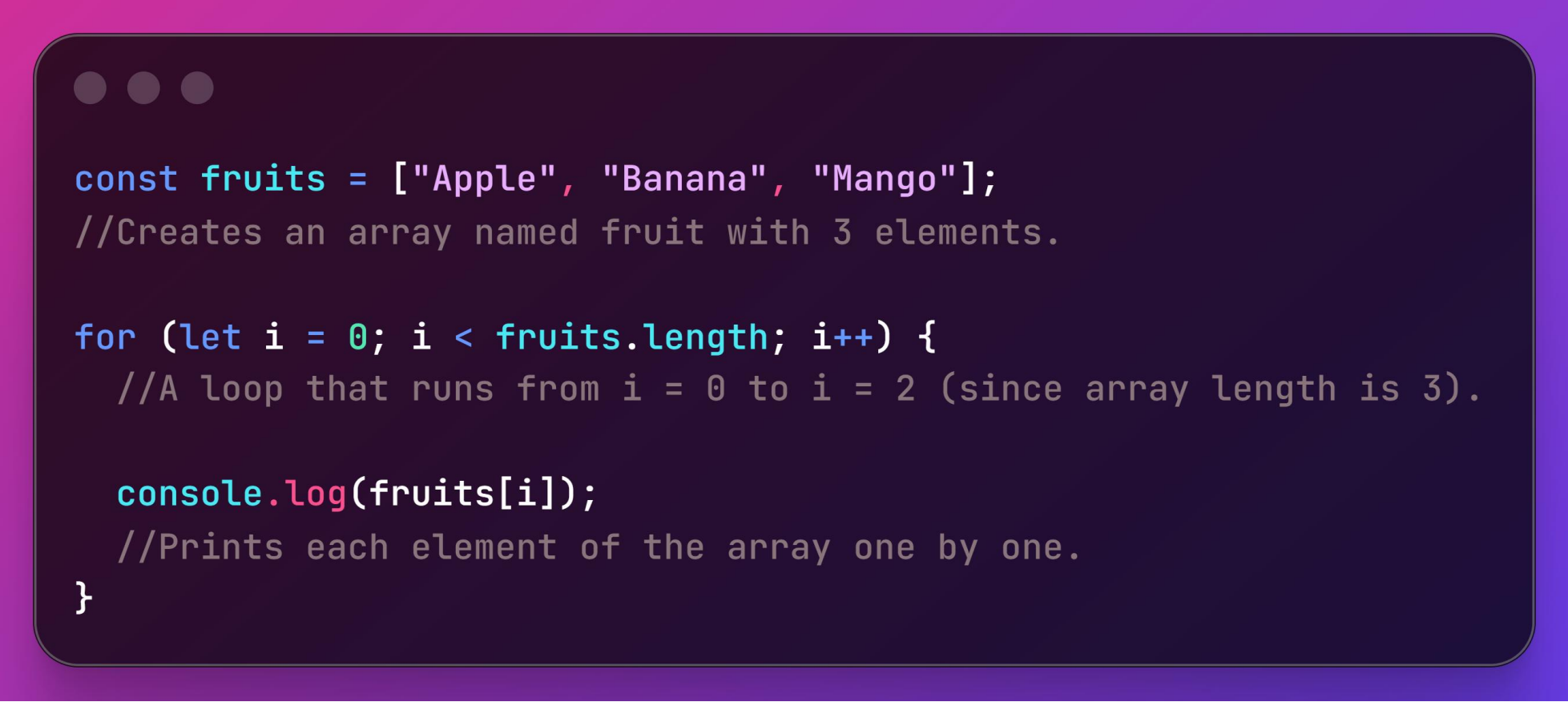
```
1  table {  
2    width: 100%;  
3    border-collapse: collapse;  
4  }  
5  th, td {  
6    border: 1px solid #ddd;  
7    padding: 8px;  
8    text-align: left;  
9  }  
10 th {  
11   background-color: #f2f2f2;  
12 }
```

ANIMATION IN CSS

```
<style>
  /* Targets the element with id="animated-box" */
  #animated-box {
    /* Applies the "spin" animation:
     - 2s = takes 2 seconds for one full cycle
     - linear = same speed throughout (no easing)
     - infinite = repeats forever */
    animation: spin 2s linear infinite;
  }

  /* Defines the animation named "spin" */
  @keyframes spin {
    /* Starts at 0 degrees (no rotation) */
    from { transform: rotate(0deg); }
    /* Ends at 360 degrees (full rotation) */
    to { transform: rotate(360deg); }
  }
</style>
```

JAVASCRIPT



```
const fruits = ["Apple", "Banana", "Mango"];  
//Creates an array named fruit with 3 elements.  
  
for (let i = 0; i < fruits.length; i++) {  
    //A loop that runs from i = 0 to i = 2 (since array length is 3).  
  
    console.log(fruits[i]);  
    //Prints each element of the array one by one.  
}
```

JAVASCRIPT CONT....

- **Code:** `document.getElementById("numberDisplay").textContent = "numbers"`
 - **Explanation:**
 - document – Refers to the whole web page.
 - .getElementById("numberDisplay") – Finds the HTML element with id="numberDisplay".
 - .textContent – Refers to the text inside that element.
 - = "numbers" – Sets that text to the word **"numbers"**.
 - **Result:**
 - The element's displayed text changes to **"numbers"**.
-

HTML CONT....

- **Bullet points:**
 - *To create bullet points in HTML, use the (unordered list) and (ordered list) tag as a container, and wrap each item inside an *
 - **Unit test:**
 - *Unit test is process to check a single module for its proper functionality. E.g. a single module of authentication, fetching data, API testing etc.*
 - **Debugging:**
 - *Debugging is a process of finding and fixing errors or bugs in computer software.*
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